

Task 9: Writing Subqueries (Nested Queries)

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox

Hints / Mini Guide:

1. Create salary data inside the employees table.
2. Write a subquery to find employees earning more than average salary.
3. Use subqueries inside WHERE, FROM, and SELECT clauses.
4. Compare subquery results with equivalent JOIN-based queries.
5. Implement correlated subqueries and observe performance differences.
6. Analyze execution flow of nested queries.
7. Identify situations where subqueries are unavoidable.
8. Practice debugging subquery errors.

Deliverables:

- SQL script containing multiple subquery examples

Final Outcome:

- Intern understands advanced query nesting and logic flow.

Interview Questions Related To Above Task:

- What is a subquery?
- Difference between correlated and non-correlated subqueries?
- Can subqueries return multiple rows?
- Subquery vs JOIN – which is better?
- Where are subqueries commonly used?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📤 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

